

Problem 1:

Let X be a topological space and let $Y \subset X$ be a subspace. Prove that $\dim Y \leq \dim X$.

Proof.

Let $Y_1 \subsetneq Y_2 \subsetneq \cdots \subsetneq Y_n$ be a max chain in Y .

Let $C_i = \overline{Y_i}$ the closure of Y_i in X .

Assume for contradiction $C_i = C_{i+1}$ this implies $Y \cap C_i = Y \cap C_{i+1} \implies Y_i = Y_{i+1}$ which is a contradiction.

Thus, $C_i \subsetneq C_{i+1}$, also since Y_i is irreducible this implies the closure is also irreducible this was shown in class.

Therefore, since $C_1 \subsetneq C_2 \subsetneq \cdots \subsetneq C_n$ is also a chain this implies $\dim Y \leq \dim X$. ■

Problem 2:

Let $R_1, \dots, R_n$ be rings. Prove that

$$\dim \left(\prod_{i=1}^n R_i \right) = \sup \{ \dim(R_i) \mid 1 \leq i \leq n \}$$

Proof.

Shown in previous homework the prime ideals in the product ring are $\prod_{i=1}^n I_i$ s.t. one $I_j = \mathfrak{p}$ s.t. $\mathfrak{p}$ is prime in R_i and I_i for $i \neq j$ is $I_i = R_i$.

Therefore, this implies that a chain in R is equivalent to a chain in R_i . Thus, the dimension of R is the max of the R_i 's if each dimension is finite and if otherwise then the dimension is infinite. Therefore, the dimension of R is the supremum of the dimensions of each R_i . ■

Problem 3:

Let k be a field.

(a)

Let $(R, \mathfrak{m})$ be a local ring which is a finite type k -algebra. Let M be an R -module of finite length. Prove that $\dim_k(R/\mathfrak{m})$ and $\dim_k(M)$ are finite, and that

$$\text{length}_R(M) \cdot \dim_k(R/\mathfrak{m}) = \dim_k(M)$$

Proof.

Shown in class since M is finite length this implies that M is noetherian. This implies that all submodules are finitely generated which implies M is finitely generated since it is a submodule.

Shown in class we know that $R/\mathfrak{m}$ is a finite extension of k . Therefore, this implies that it is a finite dimension vector space.

Let $M_0 = 0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$ be a composition series which is implies from M being finite length.

Let S be a simple R -module. This implies that $\mathfrak{m}S = S$ or $\mathfrak{m}S = 0$

Assume for contradiction that $\mathfrak{m}S = S$ by Nakayama's lemma this implies that $S = 0$ which is a contradiction of S being simple.

Therefore, $\mathfrak{m}S = 0$. This implies that $\text{Ann}_R(S) = \mathfrak{m}$ since $\text{Ann}_R(S) \neq R$ and $\mathfrak{m}$ is maximal. This implies that S is a simple $R/\mathfrak{m}$ -module. Thus, since $R/\mathfrak{m}$ is a field, and $S \neq 0$ this implies S that S must be 1 ■